



DISCRETE PROBABILITY DISTRIBUTIONS



OBJECTIVES

- Review concepts in discrete and continuous variables.
- Define what random variable is.
- Discuss what is probability distribution and discrete probability distribution.
- Demonstrate how to compute for discrete probability distribution.

DEFINITIONS

- A **discrete random** variable has either a finite number of values or a countable number of values, where “countable” refers to the fact that there might be infinitely many values, but they can be associated with a counting process.
- A **continuous random** variable has infinitely many values, and those values can be associated with measurements on a continuous scale without gaps or interruptions.



(a) Discrete Random Variable: Count of the number of movie patrons.



(b) Continuous Random Variable: The measured voltage of a smoke detector battery.

EXAMPLES- THE FOLLOWING ARE EXAMPLES OF DISCRETE AND CONTINUOUS RANDOM VARIABLES.

1. Let x the number of eggs that a hen lays in a day.

This is a discrete random variable because its only possible values are 0, or 1, or 2, and so on. No hen can lay 2.343115 eggs, which would have been possible if the data had come from a continuous scale.

■ 2. The count of the number of statistics students present in class on a given day is a whole number and is therefore a discrete random variable. The counting device shown is capable of indicating only a finite number of values, so it is used to obtain values for a discrete random variable.

EXAMPLE

3. Let x the amount of milk a cow produces in one day.

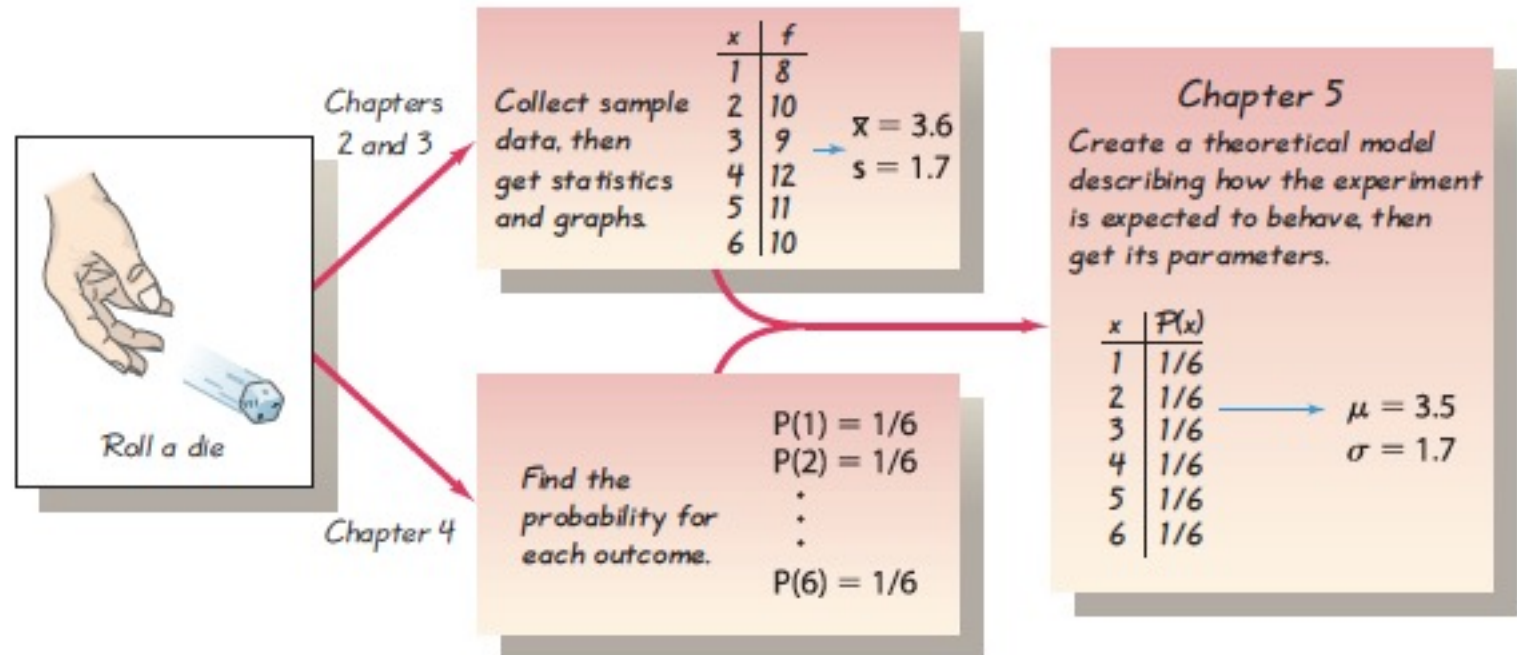
This is a continuous random variable because it can have any value over a continuous span. During a single day, a cow might yield an amount of milk that can be any value between 0 gallons and 5 gallons. It would be possible to get 4.123456 gallons, because the cow is not restricted to the discrete amounts of 0, 1, 2, 3, 4, or 5 gallons.

4. The measure of voltage for a particular smoke detector battery can be any value between 0 volts and 9 volts.

It is therefore a continuous random variable. The voltmeter as shown in the picture is capable of indicating values on a continuous scale, so it can be used to obtain values for a continuous random variable.

INTRODUCTION

- combine the methods of *descriptive statistics* and those of *probability*.



DEFINITIONS

- A *random variable* is a variable (typically represented by x) that has a single numerical value, determined by chance, for each outcome of a procedure.
- A *variable x* is a *random variable* if the value that it assumes, corresponding to the outcome of an experiment, is a chance or random event.

YOU CAN THINK OF MANY EXAMPLES OF RANDOM VARIABLES:

- x = Number of defects on a randomly selected piece of furniture
- x = SAT score for a randomly selected college applicant
- x = Number of telephone calls received by a crisis intervention hotline during a randomly selected time period



PROBABILITY DISTRIBUTIONS

- In the previous topics, you learned how to construct the relative frequency distribution for a set of numerical measurements on a variable x .

The distribution gave this information about x :

- What values of x occurred
- How often each value of x occurred

You also learned how to use the mean and standard deviation to measure the center and variability of this data set.

DEFINITIONS

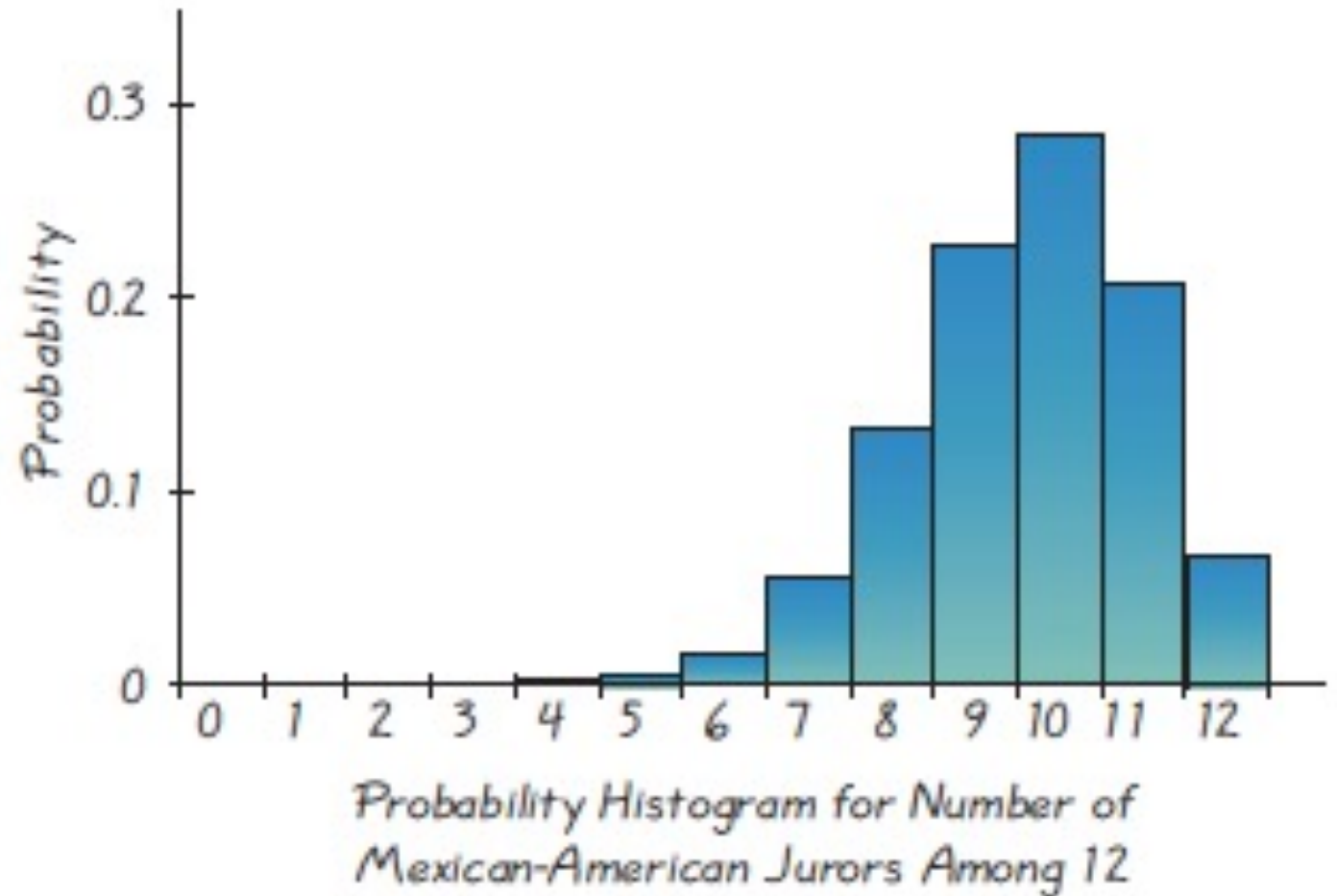
- A *probability distribution* is a description that gives the probability for each value of the random variable. It is often expressed in the format of a graph, table, or formula.
- The *probability distribution for a discrete random variable* is a formula, table, or graph that gives the possible values of x , and the probability $p(x)$ associated with each value of x .

NOTE:

- The values of x represent *mutually exclusive* numerical events.
- Summing $p(x)$ over all values of x is equivalent to adding the probabilities of all simple events and therefore equals 1.

GRAPHS

- There are various ways to graph a probability distribution, but we will consider only the probability histogram.
- A **probability histogram** is a graph that shows the **probability** of each outcome on the y - axis.



REQUIREMENTS FOR A DISCRETE PROBABILITY DISTRIBUTION

- The first requirement comes from the simple fact that the random variable x represents all possible events in the entire sample space, so we are certain (with probability 1) that one of the events will occur.

Requirements for a Probability Distribution

1. $\sum P(x) = 1$ where x assumes all possible values. (That is, the sum of all probabilities must be 1.)
2. $0 \leq P(x) \leq 1$ for every individual value of x . (That is, each probability value must be between 0 and 1 inclusive.)

REQUIREMENTS FOR A DISCRETE PROBABILITY DISTRIBUTION

- But in other cases values such as 0.999 or 1.001 are acceptable because they result from rounding errors.) Also, the probability rule stating $0 \leq P(x) \leq 1$ for any event A implies that $P(x)$ must be between 0 and 1 for any value of x .

EXAMPLES

EXAMPLE Does Table 5-2 describe a probability distribution?

SOLUTION To be a probability distribution, $P(x)$ must satisfy the preceding two requirements. But

$$\begin{aligned}\Sigma P(x) &= P(0) + P(1) + P(2) + P(3) \\ &= 0.2 + 0.5 + 0.4 + 0.3 \\ &= 1.4 \quad [\text{showing that } \Sigma P(x) \neq 1]\end{aligned}$$

Because the first requirement is not satisfied, we conclude that Table 5-2 does not describe a probability distribution.

Table 5-2

Probabilities for a
Random Variable

x	$P(x)$
0	0.2
1	0.5
2	0.4
3	0.3

EXAMPLES

EXAMPLE Does $P(x) = x/3$ (where x can be 0, 1, or 2) determine a probability distribution?

SOLUTION For the given function we find that $P(0) = 0/3$, $P(1) = 1/3$ and $P(2) = 2/3$, so that

1.
$$\sum P(x) = \frac{0}{3} + \frac{1}{3} + \frac{2}{3} = \frac{3}{3} = 1$$

2. Each of the $P(x)$ values is between 0 and 1.

Because both requirements are satisfied, the $P(x)$ function given in this example is a probability distribution.

EXAMPLES

- Toss two fair coins and let x equal the number of heads observed. Find the probability distribution for x .

Solution The simple events for this experiment with their respective probabilities are listed in Table 4.7. Since $E_1 = HH$ results in two heads, this simple event results in the value $x = 2$. Similarly, the value $x = 1$ is assigned to E_2 , and so on.

Simple Events and Probabilities in Tossing Two Coins

Simple Event	Coin 1	Coin 2	$P(E_i)$	x
E_1	H	H	1/4	2
E_2	H	T	1/4	1
E_3	T	H	1/4	1
E_4	T	T	1/4	0

EXAMPLES

For each value of x , you can calculate $p(x)$ by adding the probabilities of the simple events in that event. For example, when $x = 0$,

$$p(0) = P(E_4) = \frac{1}{4}$$

and when $x = 1$,

$$p(1) = P(E_2) + P(E_3) = \frac{1}{2}$$

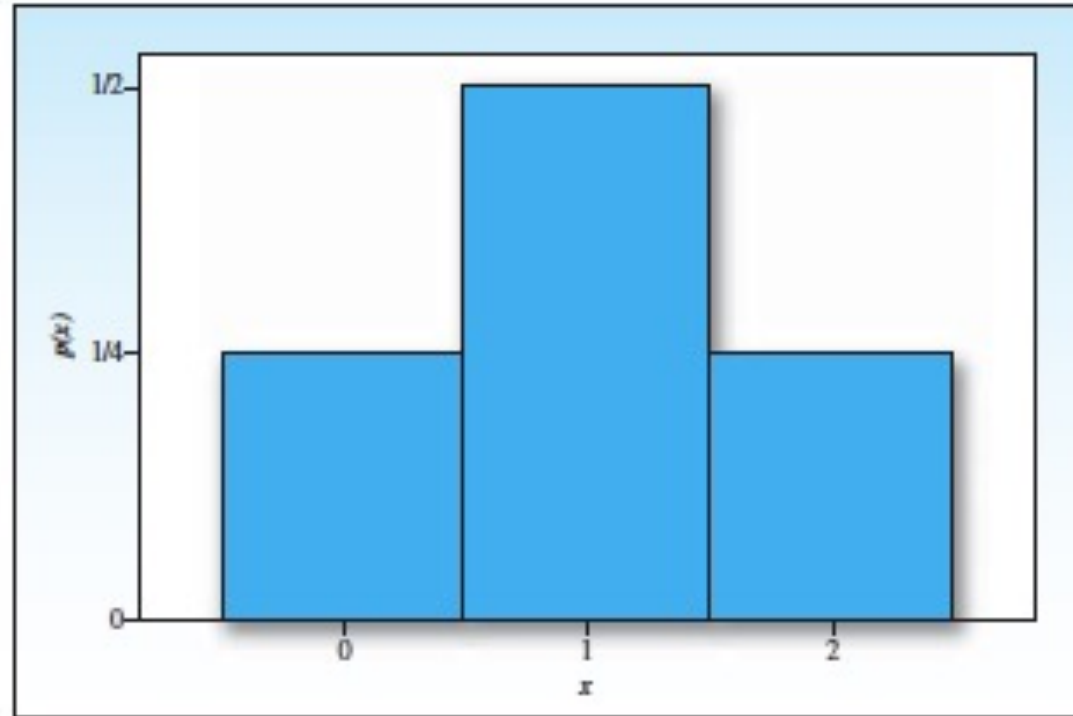
EXAMPLE

- The values of x and their respective probabilities, $p(x)$, are listed below:

Probability Distribution for x (x = Number of Heads)

Simple Events		
x	in x	$p(x)$
0	E_4	1/4
1	E_2, E_3	1/2
2	E_1	1/4
		$\Sigma p(x) = 1$

GRAPHING THE RESULTS USING PROBABILITY HISTOGRAM





THE MEAN AND STANDARD DEVIATION FOR A DISCRETE RANDOM VARIABLE



DEFINITION

Definition Let x be a discrete random variable with probability distribution $p(x)$. The mean or expected value of x is given as

$$\mu = E(x) = \sum xp(x)$$

where the elements are summed over all values of the random variable x .

DEFINITION

Definition Let x be a discrete random variable with probability distribution $p(x)$ and mean μ . The variance of x is

$$\sigma^2 = E[(x - \mu)^2] = \sum (x - \mu)^2 p(x)$$

where the summation is over all values of the random variable x .[†]

DEFINITION

Definition The standard deviation σ of a random variable x is equal to the positive square root of its variance.

EXAMPLE

- An electronics store sells a particular model of computer notebook. There are only four notebooks in stock, and the manager wonders what today's demand for this particular model will be. She learns from the marketing department that the probability distribution for x , the daily demand for the laptop, is as shown in the table.
- Find the mean, variance, and standard deviation of x . Is it likely that five or more customers will want to buy a laptop today?

x	0	1	2	3	4	5
$p(x)$	.10	.40	.20	.15	.10	.05

EXAMPLE-SOLUTION

Solution Table 4.9 shows the values of x and $p(x)$, along with the individual terms used in the formulas for μ and σ^2 . The sum of the values in the third column is

$$\mu = \sum xp(x) = (0)(.10) + (1)(.40) + \dots + (5)(.05) = 1.90$$

while the sum of the values in the fifth column is

$$\begin{aligned}\sigma^2 &= \sum (x - \mu)^2 p(x) \\ &= (0 - 1.9)^2(.10) + (1 - 1.9)^2(.40) + \dots + (5 - 1.9)^2(.05) = 1.79\end{aligned}$$

and

$$\sigma = \sqrt{\sigma^2} = \sqrt{1.79} = 1.34$$

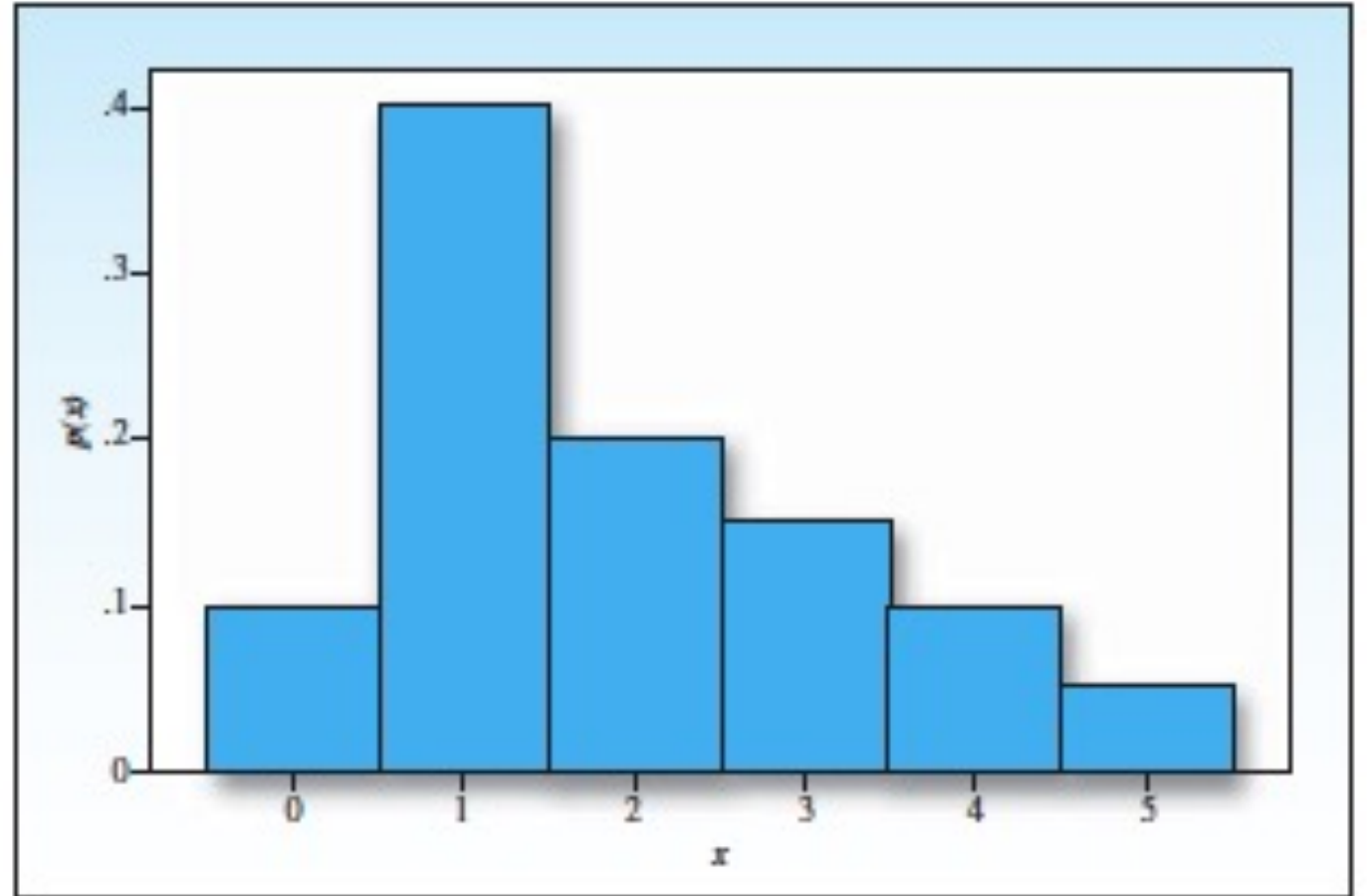
EXAMPLE-SOLUTION

Calculations for Example 4.26

x	$p(x)$	$xp(x)$	$(x - \mu)^2$	$(x - \mu)^2 p(x)$
0	.10	.00	3.61	.361
1	.40	.40	.81	.324
2	.20	.40	.01	.002
3	.15	.45	1.21	.1815
4	.10	.40	4.41	.441
5	.05	.25	9.61	.4805
Totals	1.00	$\mu = 1.90$		$\sigma^2 = 1.79$

GRAPH-PROBABILITY DISTRIBUTION

- The graph of the probability distribution is shown in the Figure.
- The graph of the probability distribution is shown in Figure 4.18. Since the distribution is approximately mound-shaped, approximately 95% of all measurements should lie within two standard deviations of the mean—that is, $\mu \pm 2\sigma \Rightarrow 1.90 \pm 2(1.34)$ or $-.78$ to 4.58



INTERPRETATION:

- Since $x = 5$ lies outside this interval, you can say it is unlikely that five or more customers will want to buy a laptop today. In fact, $P(x \geq 5)$ is exactly .05, or 1 time in 20.

PRACTICE

- A random variable x can assume five values: 0, 1, 2, 3, 4. A portion of the probability distribution is shown here:

x	0	1	2	3	4
$p(x)$	.1	.3	.3	?	.1

- Find $p(3)$.
- Construct a probability histogram for $p(x)$.
- Calculate the population mean, variance, and standard deviation.
- What is the probability that x is greater than 2?
- What is the probability that x is 3 or less?